

Indicators and Regions Acronyms

Table 1. List of acronyms in the indicators' name and regions.

Acronym	Description
Dmnl	Dimensionless
RES	Renewable Energy Source
FE	Final Energy
PE	Primary Energy
Forest M	Managed Forest
Forest P	Primary Forest
GHG	Green House Gases
C	Carbon
LUC	Land-Use Change
HDI	Human Development Index
GDP	Gross Domestic Product
EU27	European Union
UK	United Kingdom
EASOC	East-Asia & Oceania
LATAM	Latin America (but Mexico)
USMCA	US, Mexico & Canada
LROW	Rest of the World
9R	9 regions. Meaning the value can be presented for 9 global regions (EU27, UK, CHINA, EASOC, INDIA, LATAM, RUSSIA, USMCA, LROW).
1R	1 region. Meaning the value is at global level.

WILIAM Model Outcomes: Definitions

This document defines the most relevant indicators modelled in the WILIAM model, along with their units of measurement.

Table 2. List of modelled indicators with their definitions and units.

Indicator	Description	Units
Temperature change	Change in global or regional mean surface temperature relative to pre-industrial level	°C
Land use area by region	Total area allocated to different land use categories (e.g., cropland, forest, urban) for each global region	Mm ²
Meat consumption	Total amount of meat consumed per year per global region	t/year
Meat consumption per person	Total amount of meat consumed per person per year per global region	kg/(year·person)
Intake after food losses	Amount of food available for human consumption after accounting for losses in the supply chain, per year, per global region per food type	kg/(year·person)

Land products available from croplands	Total agricultural output from croplands per year per global region. Land products available are corn, rice, cereals, other cereals, soy, pulses nuts, oil crops, sugar crops, fruits vegetables, biofuels 2gcrops, other crops, wood, residues	t/year
Land products available from cropland global	Total agricultural output from croplands per year globally	t/year
Energy demand by forestry products	Energy demanded to produce products from forestry, per global region	EJ/year
Net new land requirements solar on land	Additional land area required to install new solar energy capacity (ground-mounted) per global region	km ² /year
Land area for solar	Total area currently occupied by solar energy infrastructure per global region	km ²
Land for solar demanded	Projected land area required to meet solar energy demand per global region	km ²
Urban land dispersion ratio	Ratio indicating how spread out urban land is relative to population density per global region	m ² /people
Carbon loss for deforestation	Carbon released to the atmosphere due to deforestation per global region	t/year
Share of regenerative agriculture	Proportion of cropland area managed by using regenerative agriculture practices per global region AND land product	Dmnl
Crops available all management	Total crop output per year considering all types of cropland management (industrial, traditional, low input, regenerative) per global region per land product	t/year
Share RES in FE	Share of renewable energy sources in final energy consumption per global region, per final energy type (electricity, gas, heat, hydrogen, liquid, solid biomass, solid fossil fuel)	Dmnl
Share FE RES vs total FE	Proportion of final energy consumption from renewable energy sources compared to total final energy per global region	Dmnl
Share RES in PE	Share of renewable energies in total primary energy including non-energy uses, per global region	Dmnl
Total FE including net trade	Total final energy consumption (historical) and demand (projections) including imports/exports of final energy commodities (or transformation inputs) per global region	EJ/year
PE by commodity	Domestic primary energy demand/consumption resulting from final energy demand and the current energy system set up broken down by energy commodity (e.g., coal, oil, solar)	EJ/year

RES consumption PE by region	Energy consumption of renewable energy sources in primary sources by global region	EJ/year
TO by commodity	Energy transformation output required to fulfil final energy demand per global region. TO = Final energy+Transmission losses+Storage losses+Sector energy own consumption+Flexibility technology (hydrogen, p2heat, p2gas, p2liquid). Disaggregated by seven final energy types (electricity, gas, heat, hydrogen, liquid, solid biomass, solid fossil fuel)	EJ/year
Area loss of forest M and P	Area loss of managed and primary forest, per global region	Mm ² /year
Forest primary area lost	Total area of primary (old-growth) forest lost, per global region. Positive values mean forest area is lost	km ² /year
Global GHG total energy emissions	Total greenhouse gas emissions from the global energy sector per year, disaggregated by GHG type (CO ₂ , N ₂ O, CH ₄) computed endogenously due to combustion of energy in production and final consumption	Gt/year
C emissions LUC and crop management region 9R	Carbon emissions from land-use change and crop management, per global region	Gt/year
C emissions from LUC, cropland, and grassland management global	Global carbon emissions from land-use change, cropland, and grassland management	Gt/year
Land use CO ₂ emissions forest management	CO ₂ emissions from forest management activities per year per global region (forest absorption)	t/year
GHG total energy emissions 9R	GHG emissions by global region and type of gas computed endogenously due to combustion of energy in production and final consumption	Gt/year
Total GHG emissions by GHG 1R	Total global GHG emissions, disaggregated by gas type	Gt/year
Total GHG CO ₂ eq emissions by region	Total GHG emissions in CO ₂ -equivalent per global region for all sources	GtCO ₂ e/year
Total CO ₂ e emissions per capita 1R	Total global CO ₂ -equivalent emissions per person	t/(year·person)
Total CO ₂ emissions 9R	Total CO ₂ emissions generated in all energy chain per global region	Gt/year
Total CO ₂ energy emissions 1R	Global total CO ₂ emissions from energy	Gt/year
GHG emissions per capita 9R	Per capita GHG emissions per global region and gas type	t/(year·person)
HDI 9R	Human Development Index per global region	HDI level (0-1)
Gini GDPpc regions	Inequality of GDP per capita distribution across regions, measured by the Gini	Ratio (0-1)



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	coefficient	
Government debt to GDP ratio	Share of public debt relative to the country's gross domestic product	Ratio (0-1)
Unemployment rate	Percentage of the labour force that is jobless and actively seeking work	Ratio (0-1)
PE GDP intensity	Primary energy consumption per unit of GDP, indicating energy efficiency	TJ/M\$ 2015
CO2 intensity of passenger transport 9R	CO2 emissions per passenger-kilometre in transport	g/(year·km·person)
Sea level rise by region	Regional variation in the increase of average sea level	m
Sea level rise	Global average increase in sea level	mm
pH ocean	Ocean acidity, reflecting the extent of ocean acidification	pH



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